

Implementation of the Savage-Dickey Density Ratio Test and Kullback-Leibler Divergence in BA3

BA3 Technical Documentation

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Abstract

This document provides a detailed mathematical description of the Savage-Dickey density ratio test and Kullback-Leibler divergence statistics as implemented in BA3 (BayesAss Edition 3) for testing hypotheses about migration rates between populations. We describe the theoretical foundations, computational methods including kernel density estimation with boundary correction, and the interpretation of results.

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1 Introduction

In population genetics, a fundamental question is whether migration occurs between populations. BA3 implements the Savage-Dickey density ratio test to formally test the hypothesis of zero migration between any pair of populations. Additionally, the Kullback-Leibler divergence provides a measure of information gain from prior to posterior, quantifying how much the data have informed our beliefs about migration rates.

2 Model Setup

2.1 Migration Rate Parameters

Let m_{ij} denote the per-generation migration rate from population j to population i (the proportion of individuals in population i that are immigrants from population j). The BA3 model imposes the following constraints:

$$m_{ii} \geq \frac{2}{3} \quad (\text{most individuals are non-migrants}) \quad (1)$$

$$\sum_{j \neq i} m_{ij} \leq \frac{1}{3} \quad (\text{bounded total immigration}) \quad (2)$$

2.2 Prior Distribution

The prior distribution for each off-diagonal migration rate m_{ij} (where $i \neq j$) is uniform on the interval $[0, \frac{1}{3}]$:

$$m_{ij} \sim \text{Uniform}\left(0, \frac{1}{3}\right), \quad i \neq j \quad (3)$$

The prior density function is therefore:

$$\pi(m_{ij}) = \begin{cases} 3 & \text{if } 0 \leq m_{ij} \leq \frac{1}{3} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

In particular, the prior density at zero is:

$$\pi(m_{ij} = 0) = 3 \quad (5)$$

The prior standard deviation of a $\text{Uniform}(0, \frac{1}{3})$ distribution is:

$$\sigma_{\text{prior}} = \frac{1/3}{\sqrt{12}} = \frac{1}{3\sqrt{12}} \approx 0.0962 \quad (6)$$

3 The Savage-Dickey Density Ratio Test

3.1 Theoretical Foundation

The Savage-Dickey density ratio (Dickey, 1971; Verdinelli & Wasserman, 1995) provides a method for computing Bayes factors when testing a point null hypothesis that is a special case of a more general alternative.

3.1.1 Hypothesis Framework

We test:

$$H_0 : m_{ij} = 0 \quad (\text{no migration from population } j \text{ to } i) \quad (7)$$

$$H_1 : m_{ij} > 0 \quad (\text{migration exists}) \quad (8)$$

3.1.2 Bayes Factor Definition

The Bayes factor in favor of the null hypothesis H_0 over the alternative H_1 is:

$$\text{BF}_{01} = \frac{P(\mathcal{D} \mid H_0)}{P(\mathcal{D} \mid H_1)} \quad (9)$$

where $\mathcal{D}$ denotes the observed genotype data.

3.1.3 Savage-Dickey Identity

The key insight of the Savage-Dickey method is that when H_0 specifies a point value θ_0 that lies within the parameter space of H_1 , the Bayes factor simplifies to:

$$\text{BF}_{01} = \frac{\pi(\theta_0 \mid \mathcal{D})}{\pi(\theta_0)} \quad (10)$$

where:

- $\pi(\theta_0 \mid \mathcal{D})$ is the *posterior* density evaluated at the point null value θ_0
- $\pi(\theta_0)$ is the *prior* density evaluated at the same point

For our migration rate test with $\theta_0 = 0$:

$$\boxed{\text{BF}_{01} = \frac{p(m_{ij} = 0 \mid \mathcal{D})}{p(m_{ij} = 0)} = \frac{p(m_{ij} = 0 \mid \mathcal{D})}{3}} \quad (11)$$

3.2 Interpretation of Bayes Factors

Following the Jeffreys scale (Jeffreys, 1961), we interpret Bayes factors as shown in Table 1.

$\log_{10}(\text{BF}_{01})$	BF_{01}	Evidence
> 2	> 100	Decisive for H_0 (no migration)
1 to 2	10 to 100	Strong for H_0
0.5 to 1	3.2 to 10	Substantial for H_0
0 to 0.5	1 to 3.2	Weak for H_0
-0.5 to 0	0.32 to 1	Weak for H_1 (migration exists)
-1 to -0.5	0.1 to 0.32	Substantial for H_1
-2 to -1	0.01 to 0.1	Strong for H_1
< -2	< 0.01	Decisive for H_1

Table 1: Jeffreys scale for interpreting Bayes factors

4 Posterior Density Estimation

4.1 Kernel Density Estimation

Since we have MCMC samples $\{m_{ij}^{(1)}, m_{ij}^{(2)}, \dots, m_{ij}^{(N)}\}$ from the posterior distribution, we estimate the posterior density using kernel density estimation (KDE).

For a standard kernel density estimator:

$$\hat{p}(x) = \frac{1}{Nh} \sum_{t=1}^N K\left(\frac{x - m_{ij}^{(t)}}{h}\right) \quad (12)$$

where $K(\cdot)$ is the kernel function and h is the bandwidth.

4.2 Gaussian Kernel

BA3 uses a Gaussian kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \quad (13)$$

The unnormalized form used in the implementation accumulates:

$$K_h(m) = \exp\left(-\frac{m^2}{2h^2}\right) \quad (14)$$

4.3 Boundary Correction via Reflection

Since migration rates are bounded below by zero ($m_{ij} \geq 0$), standard KDE would underestimate the density near the boundary. BA3 uses the *reflection method* (Silverman, 1986) to correct for this boundary effect.

The reflection method conceptually “reflects” each data point about the boundary, effectively doubling the contribution to the density estimate at the boundary. For estimation at $x = 0$:

$$\hat{p}(0 \mid \mathcal{D}) = \frac{2}{Nh\sqrt{2\pi}} \sum_{t=1}^N \exp\left(-\frac{(m_{ij}^{(t)})^2}{2h^2}\right) \quad (15)$$

The factor of 2 accounts for the reflection.

4.4 Bandwidth Selection

BA3 uses two bandwidth values:

4.4.1 Reference Bandwidth

During MCMC sampling, a fixed reference bandwidth is used for accumulating kernel sums:

$$h_{\text{ref}} = 0.02 \quad (16)$$

4.4.2 Silverman’s Rule of Thumb

For final density estimation, the optimal bandwidth is computed using Silverman’s rule:

$$h_{\text{opt}} = 1.06 \cdot \hat{\sigma} \cdot N^{-1/5} \quad (17)$$

where $\hat{\sigma}$ is the sample standard deviation of the posterior samples:

$$\hat{\sigma} = \sqrt{\frac{1}{N} \sum_{t=1}^N (m_{ij}^{(t)})^2 - \left(\frac{1}{N} \sum_{t=1}^N m_{ij}^{(t)} \right)^2} \quad (18)$$

A minimum bandwidth of $h_{\text{min}} = 0.01$ is enforced to prevent numerical instability.

4.5 Histogram-Based Fallback

As a sanity check, BA3 also computes a histogram-based density estimate:

$$\hat{p}_{\text{hist}}(0 \mid \mathcal{D}) = \frac{n_{\text{near zero}}}{N \cdot \epsilon} \quad (19)$$

where:

- $n_{\text{near zero}}$ is the count of samples with $m_{ij} < \epsilon$
- $\epsilon = 0.005$ is the threshold for “near zero”

If the KDE estimate differs from the histogram estimate by more than a factor of 100, the histogram estimate is used as a fallback.

4.6 Complete Posterior Density Estimate

The final posterior density at zero is:

$$\hat{p}(0 \mid \mathcal{D}) = \frac{2}{N h_{\text{opt}} \sqrt{2\pi}} \sum_{t=1}^N \exp \left(-\frac{(m_{ij}^{(t)})^2}{2 h_{\text{ref}}^2} \right) \quad (20)$$

Note: The implementation accumulates kernel values using h_{ref} during MCMC but applies normalization using h_{opt} at the end.

5 Kullback-Leibler Divergence

5.1 Definition

The Kullback-Leibler (KL) divergence measures the information gained when updating from the prior distribution to the posterior distribution:

$$\text{KL}(p_{\text{post}} \parallel p_{\text{prior}}) = \int p(m_{ij} \mid \mathcal{D}) \log \frac{p(m_{ij} \mid \mathcal{D})}{p(m_{ij})} dm_{ij} \quad (21)$$

This quantity is also known as the *information gain* or *relative entropy*.

5.2 Hybrid Analytical-Numerical Approach

A Gaussian approximation for KL divergence is inappropriate here because:

1. The prior is $\text{Uniform}(0, \frac{1}{3})$ with bounded support, not Gaussian
2. The posterior is also bounded on $[0, \frac{1}{3}]$
3. Posteriors for migration rates are often highly skewed toward zero

Instead, BA3 uses the **exact analytical prior density** combined with **kernel density estimation** for the posterior.

5.2.1 Analytical Prior

The prior density for off-diagonal migration rates is:

$$p(m_{ij}) = \begin{cases} 3 & \text{if } m_{ij} \in [0, \frac{1}{3}] \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

5.2.2 KDE for Posterior

The posterior density is estimated using kernel density estimation with a Gaussian kernel and reflection boundary correction (as described in Section 4):

$$\hat{p}(m \mid \mathcal{D}) = \frac{1}{Nh} \sum_{t=1}^N \left[K\left(\frac{m - m^{(t)}}{h}\right) + K\left(\frac{m + m^{(t)}}{h}\right) \right] \quad (23)$$

where $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ is the Gaussian kernel and the second term provides reflection at the boundary $m = 0$.

5.2.3 Simplification with Uniform Prior

Substituting the uniform prior into the KL divergence:

$$\text{KL}(\text{post} \parallel \text{prior}) = \int_0^{1/3} \hat{p}(m \mid \mathcal{D}) \log \frac{\hat{p}(m \mid \mathcal{D})}{3} dm \quad (24)$$

$$= \int_0^{1/3} \hat{p}(m \mid \mathcal{D}) \log \hat{p}(m \mid \mathcal{D}) dm - \int_0^{1/3} \hat{p}(m \mid \mathcal{D}) \log 3 dm \quad (25)$$

$$= \int_0^{1/3} \hat{p}(m \mid \mathcal{D}) \log \hat{p}(m \mid \mathcal{D}) dm - \log 3 \quad (26)$$

Since the posterior integrates to 1 over its support. Recognizing that the first term is the negative differential entropy:

$$\boxed{\text{KL}(\text{post} \parallel \text{prior}) = -H(\text{posterior}) - \log 3} \quad (27)$$

where

$$H(\text{posterior}) = - \int_0^{1/3} \hat{p}(m \mid \mathcal{D}) \log \hat{p}(m \mid \mathcal{D}) dm \quad (28)$$

5.3 Numerical Integration

The integral is computed numerically using the trapezoidal rule over a fine grid on $[0, \frac{1}{3}]$:

$$\text{KL} \approx \sum_{i=0}^{G-1} \hat{p}(m_i | \mathcal{D}) \log \frac{\hat{p}(m_i | \mathcal{D})}{3} \cdot \Delta m \quad (29)$$

where:

- G is the number of grid points (default: 1000)
- $\Delta m = \frac{1}{3G}$ is the grid spacing
- $m_i = i \cdot \Delta m$ for $i = 0, 1, \dots, G - 1$

5.3.1 Handling Zero Density

When $\hat{p}(m_i | \mathcal{D}) = 0$, the integrand $p \log p$ is defined as 0 (by continuity, since $\lim_{p \rightarrow 0^+} p \log p = 0$). In practice, we skip grid points where the density falls below a small threshold $\epsilon_{\text{floor}} = 10^{-10}$.

5.4 Conversion to Bits

The KL divergence is reported in bits (base-2 logarithm) rather than nats:

$$\text{KL}_{\text{bits}} = \frac{\text{KL}_{\text{nats}}}{\ln 2} = \text{KL}_{\text{nats}} \cdot \log_2 e \quad (30)$$

where $\log_2 e \approx 1.4427$.

5.5 Interpretation

The KL divergence in bits can be interpreted as the expected number of additional bits needed to encode samples from the posterior using a code optimized for the prior:

- **0 bits:** No information gained; posterior equals prior
- **1 bit:** Data provide moderate information about migration rate
- **2+ bits:** Data are highly informative; posterior is much more concentrated than prior
- **Large values:** Strong evidence that migration rate is constrained to a narrow range

Note that the maximum possible KL divergence for a uniform prior on $[0, \frac{1}{3}]$ is unbounded above (approached as the posterior concentrates to a point mass).

6 Algorithm Summary

Algorithm 1 Savage-Dickey Test and KL Divergence Computation

```

1: Initialize: kernelSum  $\leftarrow$  0, sumM  $\leftarrow$  0, sumM2  $\leftarrow$  0,  $n_{\text{near}} \leftarrow$  0,  $N \leftarrow$  0
2: Set constants:  $h_{\text{ref}} \leftarrow$  0.02,  $\epsilon \leftarrow$  0.005,  $\pi(0) \leftarrow$  3,  $G \leftarrow$  1000
3: Initialize sample storage: samples  $\leftarrow$  empty list
4: for each MCMC sample  $m$  (post burn-in) do
5:   sumM  $\leftarrow$  sumM +  $m$ 
6:   sumM2  $\leftarrow$  sumM2 +  $m^2$ 
7:   kernelSum  $\leftarrow$  kernelSum +  $\exp\left(-\frac{m^2}{2h_{\text{ref}}^2}\right)$ 
8:   Append  $m$  to samples
9:   if  $m < \epsilon$  then
10:      $n_{\text{near}} \leftarrow n_{\text{near}} + 1$ 
11:   end if
12:    $N \leftarrow N + 1$ 
13: end for
14:  $\hat{\mu} \leftarrow$  sumM/ $N$ 
15:  $\hat{\sigma} \leftarrow \sqrt{\text{sumM2}/N - \hat{\mu}^2}$ 
16:  $h_{\text{opt}} \leftarrow \max(0.01, 1.06 \cdot \hat{\sigma} \cdot N^{-0.2})$ 
17: ▷ Savage-Dickey Bayes Factor
18:  $\hat{p}(0|\mathcal{D}) \leftarrow \frac{2 \cdot \text{kernelSum}}{N \cdot h_{\text{opt}} \cdot \sqrt{2\pi}}$ 
19:  $\hat{p}_{\text{hist}} \leftarrow n_{\text{near}} / (N \cdot \epsilon)$ 
20: if  $\hat{p}(0|\mathcal{D}) < 0.01 \cdot \hat{p}_{\text{hist}}$  or  $\hat{p}(0|\mathcal{D}) > 100 \cdot \hat{p}_{\text{hist}}$  then
21:    $\hat{p}(0|\mathcal{D}) \leftarrow \hat{p}_{\text{hist}}$  ▷ Fallback to histogram
22: end if
23:  $\text{BF}_{01} \leftarrow \hat{p}(0|\mathcal{D}) / \pi(0)$ 
24: ▷ KL Divergence via Numerical Integration
25:  $\Delta m \leftarrow \frac{1}{3G}$ 
26:  $\text{KL} \leftarrow 0$ 
27: for  $i = 0$  to  $G - 1$  do
28:    $m_i \leftarrow i \cdot \Delta m$ 
29:    $\hat{p}_i \leftarrow \frac{1}{Nh_{\text{opt}}} \sum_{t=1}^N \left[ K\left(\frac{m_i - \text{samples}[t]}{h_{\text{opt}}}\right) + K\left(\frac{m_i + \text{samples}[t]}{h_{\text{opt}}}\right) \right]$ 
30:   if  $\hat{p}_i > 10^{-10}$  then
31:      $\text{KL} \leftarrow \text{KL} + \hat{p}_i \cdot \log(\hat{p}_i/3) \cdot \Delta m$ 
32:   end if
33: end for
34:  $\text{KL}_{\text{bits}} \leftarrow \text{KL} \cdot \log_2 e$ 
35: return  $\text{BF}_{01}$ ,  $\log_{10}(\text{BF}_{01})$ ,  $\text{KL}_{\text{bits}}$ 

```

7 Implementation Details

7.1 Data Structure

The statistics are accumulated in the `SavageDickeyStats` structure:

```

struct SavageDickeyStats {
    double kernelSum;           // Sum of kernel evaluations at zero
    double sumM;               // Running sum of m (for mean)
    double sumM2;              // Running sum of m^2 (for variance)
    long int countNearZero;    // Count where m < 0.005

```

```

    long int nSamples;           // Total sample count
    std::vector<double> samples; // Stored samples for KL computation
};

```

7.2 Online vs. Stored Computation

For the Savage-Dickey test, statistics are updated incrementally using Welford-style online algorithms:

$$\bar{m}_N = \frac{1}{N} \sum_{t=1}^N m^{(t)} = \frac{\text{sumM}}{N} \quad (31)$$

$$\hat{\sigma}_N^2 = \frac{\text{sumM2}}{N} - \left(\frac{\text{sumM}}{N} \right)^2 \quad (32)$$

For the KL divergence, samples must be stored to evaluate the KDE at arbitrary grid points during numerical integration. The memory requirement is $O(N \cdot P^2)$ where N is the number of post-burn-in samples and P is the number of populations. For typical analyses ($N \approx 2500$ samples, $P \leq 10$ populations), this requires approximately 2 MB of storage.

8 Limitations and Considerations

8.1 Bandwidth Sensitivity

The Savage-Dickey test relies on accurate density estimation at a single point (zero). This estimate is sensitive to:

- Bandwidth selection
- Number of MCMC samples
- Posterior concentration near the boundary

8.2 Boundary Effects

The reflection method assumes the density is symmetric about the boundary, which may not hold for all posteriors. When the posterior mass is concentrated away from zero, this has minimal impact.

8.3 KL Divergence Numerical Integration

The numerical integration for KL divergence depends on:

- **Grid resolution:** Finer grids improve accuracy but increase computation time. The default of 1000 grid points provides a good balance.
- **KDE accuracy:** The quality of the posterior density estimate affects the KL divergence. Sufficient MCMC samples (typically >1000) are needed.
- **Density floor:** Very small density values can cause numerical instability in $\log(\hat{p})$. Values below 10^{-10} are treated as zero.

The hybrid approach using the exact uniform prior avoids the Gaussian approximation errors that would occur with bounded, skewed posteriors.

9 References

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